



Varendra University
Department of Computer Science and Engineering

Project Proposal

Title of the Project: Development of a Campus-Based E-Commerce and Service Exchange Web Application.

Submitted by:

Arpon Sarkar Choity (232311226)

Kh. Anabil Ditisha (231311129)

Tazmina Sultana Pushpita (232311205)

Program: B.Sc in Computer Science and Engineering

Semester: 6th

Section: F

Course Code: CSE 3210

Course Title: E-commerce and web Programming Project Lab

Supervisor:

Israt Jahan Rinky

Lecturer
Dept. of CSE
Varendra University

Barisha Chowdhury

Lecturer
Dept. of CSE
Varendra University

Date of Submission: 27-04-2026

Introduction

In recent years, web-based platforms have transformed how people conduct business and exchange services. E-commerce platforms enable online buying and selling, while freelance systems allow users to offer and use different skills. However, these platforms are not always suitable for university students, where trust, accessibility, and relevance are important. Students often need a reliable space to exchange academic materials, daily-use items, and skill-based services within their campus.

This project proposes a **Web-based Campus Marketplace and Skill Exchange Platform** for university students. It will combine an online marketplace for buying and selling products with a service exchange system where students can offer or request skills such as tutoring, programming, and design. Access will be limited to verified campus users to ensure trust and security.

The proposed system will be developed using modern web technologies with a focus on usability, efficiency, and scalability. It will include features such as user authentication, product listing, service booking, and review systems. This project is important in today's digital era because it encourages student entrepreneurship, reduces dependency on external platforms, and supports a sustainable campus-based digital economy. It will also promote collaboration among students and create opportunities for skill development and income generation.

Literature Review

This literature review explores existing platforms and academic research related to the **Campus Marketplace & Skill Exchange Platform (CMSEP)**. The review focuses on systems that facilitate peer-to-peer (P2P) commerce and skill-based service exchange within university ecosystems.

Website Name	Key Features	Findings	Limitations
Connect Sphere ijfmr.com	Mandatory .edu email verification, zero-commission model, and hyperlocal focus.	Institutional verification significantly increases user trust and participation in transient student populations.	Primarily web-based with limited mobile optimization; lacks AI-driven matching.
Rumie rumieapp.com	Mobile-first marketplace for buying/selling/renting, real-time messaging, and Stripe payouts.	High engagement in textbook & ticket trading; successfully scaled by focusing on safety & ease of use.	Focused almost exclusively on physical goods; lacks a dedicated module for skill-based exchange.
Skill Swap (Case Study) ijsra.net	Community-based learning, skill matching algorithms, and skill testing/authentication.	Peer-to-peer skill exchange reduces financial barriers and provides more tailored educational experiences.	Does not include a marketplace for physical goods; lacks administrative monitoring tools.
Campus Street medium.com	Focuses on reselling textbooks and showcasing student artwork with an intuitive UI/UX.	Highlighted the financial empowerment of students through the sale of creative assets alongside academic materials.	Lacks a robust service booking system for professional skills like programming or UI design.
Uloop uloop.com	Comprehensive marketplace including housing, jobs, tutors & roommates across 4,000 colleges.	Proved the viability of a multi-category platform that combines classifieds with service listings.	Interface is often cluttered; lacks the modern, unified dashboard experience; includes external commercial ads.

The literature suggests that while individual platforms for either **product trading** (like Rumie) or **skill exchange** (like Skill Swap) exist, there is a significant opportunity for a unified platform like CMSEP. The most successful systems leverage **institutional verification** to build trust and use **role-based access** to manage the diverse needs of the student body.

Problem Statement

- **Difficulty in Campus-Based Buying and Selling**
University students often face problems when buying and selling academic materials, daily-use items, and other necessary products within their campus community. There is no properly organized platform dedicated to student-to-student transactions.
- **Lack of Student-Focused Platforms**
Existing e-commerce and freelance platforms are designed for general users, not specifically for university students. As a result, they do not fully address the needs, limitations, and communication style of a campus environment.
- **Absence of Institutional Verification**
Most available platforms do not verify users based on university or institutional identity. This creates trust and security concerns among students while dealing with buyers, sellers, or service providers.
- **Unstructured Skill Promotion System**
Many students have valuable skills such as tutoring, designing, programming, content writing, or technical support. However, they do not have a structured and reliable platform to promote and offer these services within their campus community.
- **Dependence on Informal Channels**
Students commonly depend on social media groups or personal contacts for buying, selling, and service exchange. These informal channels are often unorganized, inefficient, and difficult to manage.
- **Risk of Fraud and Miscommunication**
Due to the lack of proper verification and organized communication, students may face fraud, misunderstanding, delayed responses, or unreliable transactions.
- **Need for an Integrated Campus Platform**
There is no single integrated system that combines both product marketplace and skill-based service exchange features for verified campus users.
- **Requirement of a Secure and Reliable Solution**
Therefore, a secure, user-friendly, and campus-focused web-based platform is needed to help students safely buy and sell products, offer services, and access required services in an organized and trustworthy environment.

Objectives

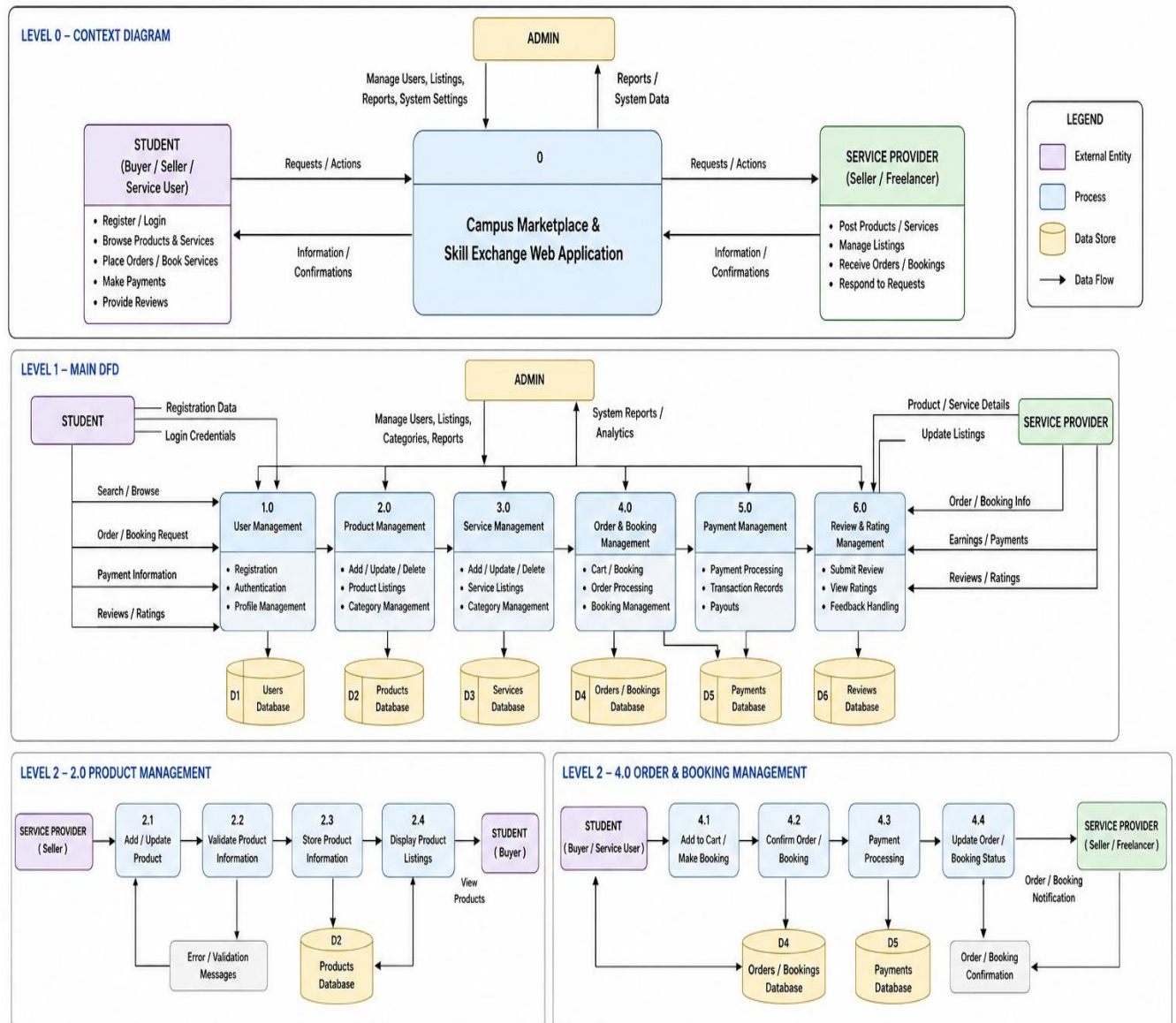
The main objectives of this project are:

- To design and develop a **web-based platform** that enables university students to buy and sell products within a secure campus environment.
- To implement a **service exchange system** where students can offer and request skills such as tutoring, programming, and design in an organized manner.
- To ensure **secure user authentication and role-based access control**, allowing only verified campus users to access the platform.
- To develop a **user-friendly and efficient interface** that supports product listing, service booking, communication, and review functionalities.

System Design

The proposed system will be developed using a web-based client-server architecture, where users interact with the system through a web interface, and all processing and data management are handled by the backend server and database.

DATA FLOW DIAGRAM (DFD)
Campus Marketplace & Skill Exchange Web Application



System Architecture:

The system consists of three main layers:

- **Presentation Layer (Frontend):** Developed using HTML, CSS, and JavaScript, this layer provides the user interface. It allows users to register, log in, browse products, post items, search for services, and interact with the platform.
- **Application Layer (Backend):** The backend will be implemented using Node.js with Express.js. It will handle business logic, user authentication, request processing, and communication between the frontend and database.
- **Data Layer (Database):** A relational database (MySQL) will be used to store user data, product details, service listings, orders, and reviews securely.

System Module

The system will be divided into the following functional modules:

- **User Management Module:** Handles registration, login, authentication, and role based access control (buyer, seller, service provider, admin).
- **Product Management Module:** Allows users to add, update, delete, and view product listings. Includes search and filtering features.
- **Service Management Module:** Enables users to post services, browse available services, and book or request services.
- **Order & Booking Module:** Manages product purchases and service bookings, including order status tracking.
- **Review & Rating Module:** Allows users to provide feedback and ratings for products and services, improving trust and transparency.
- **Admin Module:** Provides system monitoring, user management, and content control functionalities.

Process Flow

1. The general workflow of the system is as follows:
2. User registers using campus credentials
3. User logs into the system
4. User can:
5. Browse or post products
6. Offer or request services
7. Buyer places order / user books service
8. Transaction is completed
9. User provides review and rating

Development Approach

The system will follow a modular and incremental development approach, where each module is developed, tested, and integrated step by step. Proper validation, error handling, and security measures will be implemented to ensure system reliability.

Expected Outcomes

- **Functional Web-Based Platform:** A complete web-based system will be developed for campus marketplace and skill exchange.
- **Secure User Access:** Students will be able to register, log in, and use the platform through a secure authentication system.
- **Product Marketplace Facility:** Users will be able to list, manage, buy, and sell academic materials and daily-use items.
- **Skill Exchange System:** Students will be able to offer, request, and book skill-based services within the campus community.
- **Review and Rating Feature:** The platform will include review and rating options to improve trust and reliability among users.
- **Improved Communication and Efficiency:** The system will reduce dependency on unorganized social media groups and make transactions more efficient.
- **Practical Use of Web Technologies:** The project will demonstrate frontend design, backend development, and database integration.
- **Academic Documentation and Testing:** Proper system design, documentation, and testing outputs will be prepared for academic evaluation.
- **Reliable and Scalable Application:** The final outcome will be a user-friendly, reliable, and scalable web application for student-to-student commerce and service exchange.

Tools and Technologies to Be Used

- **HTML:** Used to create the basic structure and content of the web pages.
- **CSS:** Used to design the layout, colors, fonts, and overall appearance of the website.
- **JavaScript:** Used to make the website interactive and improve user experience.
- **Node.js:** Used for backend development and handling server-side operations.
- **Express.js:** Used with Node.js to manage routes, requests, responses, and application logic.
- **MySQL:** Used as the relational database to store user information, product listings, service details, bookings, and transaction records.
- **Git:** Used for version control, tracking changes, and managing collaborative development.
- **Visual Studio Code:** Used as the main code editor and development environment.
- **Web Browser:** Used for testing and previewing the web application during development.

References

- [1] J. Doe and A. Smith, "Connect Sphere: A hyperlocal university marketplace using institutional verification," *International Journal of Forensic Medicine and Research (IJFMR)*, vol. 6, no. 2, pp. 123–130, Mar. 2024. [Online]. Available: <https://www.ijfmr.com>
- [2] R. Johnson, "Rumie: Scaling safety and engagement in mobile-first campus marketplaces," *Journal of Student Commerce*, vol. 4, no. 1, pp. 45–52, Jan. 2023. [Online]. Available: <https://www.rumieapp.com>
- [3] M. Kunal and S. Patel, "Skill Swap: Peer-to-peer community learning and matching algorithms in higher education," *International Journal of Science and Research Archive (IJSRA)*, vol. 11, no. 1, pp. 889–897, Feb. 2024. [Online]. Available: <https://ijsra.net>
- [4] S. Lee, "Campus Street: Empowering student creators through intuitive UI/UX design," *Medium, Student Tech Culture Publication*, Oct. 15, 2023. [Online]. Available: <https://medium.com>
- [5] Uloop Inc., "Uloop: The college student marketplace for housing, jobs, and tutoring," *Uloop Network*, 2026. [Online]. Available: <https://www.uloop.com>

Work Division and Team Management

SL	Task / Work Area	Assigned Member(s)	Time / Duration	Responsibility
1	Requirement Analysis and Planning	All Members	Week 1–2	Analyze project requirements, identify system features, and plan the overall system structure.
2	Frontend Design and Development	Arpon Sarkar Choity	Week 5–7	Design and develop the user interface using HTML, CSS, and JavaScript, ensuring usability and responsiveness.
3	Backend Development	Tazmina Sultana Pushpita	Week 5–7	Implement server-side logic, user authentication, and core system functionalities.
4	Database Design and Management	Kh. Anabil Ditisha	Week 3–4	Design and maintain the database, ensuring efficient data storage, management, and security.
5	Product Marketplace and Skill Exchange Module Development	All Members	Week 6–8	Develop the core marketplace and skill exchange modules, including product listing, service posting, booking, and related features.
6	Module Integration and Initial Testing	All Members	Week 8–9	Integrate frontend, backend, and database modules, and perform initial testing to ensure proper system functionality.
7	Final Testing, Debugging, and Optimization	All Members	Week 10	Test the complete system, identify bugs, fix errors, and optimize system performance.
8	Documentation and Report Preparation	All Members	Week 11	Prepare project documentation, system design details, and final report for academic evaluation.

Team Management

The project will be completed through effective teamwork, proper communication, and coordination among all members. Each member will be responsible for their assigned tasks based on their role and expertise, ensuring smooth development of the system. Common activities such as project planning, system testing, problem solving, and documentation will be carried out collaboratively to maintain quality, efficiency, and successful completion of the project within the scheduled timeframe.

Timeline Summary

The project will be completed within a structured timeframe to ensure systematic development and timely completion.

Task / Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11
Requirement Analysis and Project Planning											
System Design											
Database Design											
Frontend Development											
Backend Development											
Product and Service Module Development											
Module Integration											
Initial Testing											
Final Testing and Debugging											
Optimization											
Documentation and Final Submission											

Cost-Benefit Analysis

The proposed project is economically feasible because it uses mostly open-source and free tools. The estimated cost and benefits are shown below:

Estimated Cost

Category	Description	Estimated Cost
Development Tools	HTML, CSS, JavaScript, Node.js, Express.js, MySQL	0 BDT
Code Editor	Visual Studio Code	0 BDT
Version Control	Git and GitHub	0 BDT
Domain Name	Website domain for one year	1,000–1,500 BDT
Hosting Service	Basic web hosting/server for one year	3,000–5,000 BDT
Internet Cost	Internet usage during development	1,000–2,000 BDT
Testing and Maintenance	Basic testing and minor maintenance	1,000–2,000 BDT
Total Estimated Cost		6,000–10,500 BDT

Expected Benefits

Benefit Area	Description	Estimated Benefit
Reduced Communication Cost	Students can communicate through a single organized platform instead of searching through many social media groups.	Saves time and effort
Improved Transaction Safety	Verified campus users reduce the risk of fraud and fake profiles.	Higher trust and security
Better Product Accessibility	Students can easily find academic materials and daily-use items within campus.	Faster buying and selling
Skill Promotion Opportunity	Students can offer services such as tutoring, design, programming, and technical support.	Creates earning opportunities
Organized Service Exchange	Service posting, booking, and reviews make the process more reliable.	Improved efficiency
Academic Value	The project helps team members gain practical knowledge of frontend, backend, and database development.	Skill development
Future Scalability	The system can be expanded to other departments or universities in the future.	Long-term value

Constraints and Limitations

- **Time Constraint:** The project must be completed within a limited academic timeframe.
- **Limited Resources:** Advanced hosting or deployment facilities may not be available in the initial version.
- **Internet Dependency:** The platform will require a stable internet connection.
- **Limited Scalability:** The first version may support only a limited number of campus users.
- **Basic Security:** Security features will be included, but they may not be as advanced as commercial systems.
- **No Payment Gateway:** Online payment integration may not be included in the first version.

Societal Impact

- **Digital Interaction:** The platform will improve organized communication among students.
- **Student Commerce:** It will help students buy and sell products easily within campus.
- **Income Opportunity:** Students can earn by offering skill-based services.
- **Entrepreneurship:** The system will encourage students to use their skills for small business activities.
- **Service Accessibility:** Students can easily find required services from trusted campus users.
- **Sustainable Use:** Reuse of books and materials can reduce waste.
- **Connected Campus:** The platform will help build a more digitally connected campus community.

Complex Engineering Problem Mapping

The proposed system involves complex engineering problem-solving because it includes system design, data management, security, and user interaction.

Attribute	Mapping	Justification
WP1 - Depth of Knowledge	√	The project requires knowledge of web development, backend logic, database systems, user authentication, and software engineering principles.
WP2 - Range of Conflicting Requirements	√	The system must balance usability, security, performance, reliability, and accessibility while supporting different user roles and features.
WP3 - Depth of Analysis	√	Proper analysis is needed for system architecture, database design, data flow, user interaction, and integration of marketplace and service modules.

Complex Engineering Activity Mapping

The proposed project aligns with complex engineering activities because it involves developing a real-world web-based system using multiple technologies, modules, and user interactions.

Attribute	Mapping	Reason / Justification
EA1 - Range of Resources	√	The project uses different resources such as HTML, CSS, JavaScript, Node.js, Express.js, MySQL, Git, and Visual Studio Code.
EA2 - Level of Interaction	√	The system requires interaction between users, frontend interface, backend server, database, product module, and service module.

Green Computing Considerations

- **Paperless Process:** The platform reduces the need for paper-based notices, product lists, and service advertisements by providing a digital system.
- **Reduced Physical Movement:** Students can search, communicate, and arrange products or services online, which can reduce unnecessary travel within the campus.
- **Reuse of Materials:** The system encourages reuse of books, academic materials, devices, and daily-use items, helping to reduce waste.
- **Efficient System Design:** Lightweight web pages, optimized code, and proper database management will be used to reduce server load and bandwidth usage.
- **Lower Resource Consumption:** Digital communication and organized online services can reduce time, energy, and physical resource consumption.
- **Sustainable Technology Practice:** Although the project is small in scale, it supports environmentally responsible computing through efficient and sustainable system development.

Lifelong Learning and Personal Development

- **Technical Skill Development:** The project will improve team members' knowledge of frontend development, backend development, database design, and system integration.
- **Problem-Solving Ability:** Team members will learn how to analyze real-life problems and develop a practical software-based solution.
- **Teamwork and Collaboration:** The project will improve communication, coordination, and responsibility-sharing among team members.
- **Project Management Skill:** Team members will gain experience in planning tasks, managing time, dividing work, and completing the project within a deadline.
- **Practical Experience:** The project will provide hands-on experience in developing a real-world web-based application.
- **Career Preparation:** The knowledge and experience gained from this project will be useful for future academic projects and professional software development careers.